Database Schemas as Quantum Fields: Companion Report — Proofs, Instruments, and Pre-registered Experiments

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Code and data: https://github.com/stevezhai139/qfdb-reproducibility

Abstract

This report accompanies the paper *Database Schemas as Quantum Fields: Relational and Document Models as a Classical Sector*. It contains the proofs of the paper's propositions, the full algebra of the worked temporal example (Faces A and B), the complementarity reading, the complete S1 elicitation design with power analysis, the pre-registered stored-data negative control with full results, and the LLM-respondent pilot harness. Every number is reproduced by the companion code.

1 Proofs

1.1 Proposition 1 (Structure vs. state)

Write an entity state as $\rho = D + C$ in the structure basis $\{|s_k\rangle\}$, with $D = \sum_k p_k |s_k\rangle\langle s_k|$ and $C = \rho - D$.

(a) The sector $\{C = 0\}$ is closed under structure change. A definite structure is a basis vector $|s_k\rangle\langle s_k|$; a heterogeneous or merely uncertain one is a mixture $\sum_k p_k|s_k\rangle\langle s_k|$. Both are diagonal, so $C = 0$. Schema evolution acts as a stochastic matrix M on the population vector p ($p \mapsto Mp$, $M_{jk} \geq 0$, $\sum_j M_{jk} = 1$): it maps diagonal states to diagonal states and leaves C untouched. Hence no sequence of structure changes creates coherence. □

(b) Which signature is carried by what. For a query outcome $|e\rangle$ the Born rule expands as

$$P(e) = \sum_k p_k \, |\langle e|s_k\rangle|^2 \; + \; 2\,\mathrm{Re}\sum_{k<l} C_{kl} \, \langle e|s_k\rangle\langle s_l|e\rangle.$$

The second (interference) term is linear in C and vanishes iff $C = 0$: the S3 signature is carried by coherence alone. Every separable ρ admits a local-hidden-variable model for the CHSH settings and therefore obeys $|S| \leq 2$ (Section 1.5): the S1 signature is carried by non-separability alone.

Order effects, by contrast, are a property of the *query pair*, not of C:

Lemma 1 (Commuting queries are order-invariant on every state)**.** Let $\{P_a\}$ and $\{Q_b\}$ be projective measurements with $[P_a, Q_b] = 0$ for all a, b. Then for every state ρ the marginal of either measurement is unchanged by performing the other first.

Proof. $\Pr(b \text{ after } P) = \sum_a \operatorname{tr}\big(Q_b P_a \rho P_a\big) = \sum_a \operatorname{tr}\big(P_a Q_b P_a \rho\big) = \sum_a \operatorname{tr}\big(Q_b P_a \rho\big) = \operatorname{tr}(Q_b \rho)$, using cyclicity and $P_a Q_b P_a = Q_b P_a^2 = Q_b P_a$. The symmetric statement holds with the roles exchanged. □

Conversely, an order effect requires non-commuting queries, and it can occur on a *coherence-free* state: Face A of the paper realises exactly this. Take $\rho = |0\rangle\langle 0|$ (a stored, definite record; $C = 0$ in the stored basis) and a second query Q_θ projecting at angle θ to the stored axis. Then

$$p_{AB}(0,\theta) = \cos^2\theta \qquad \text{but} \qquad p_{BA}(0,\theta) = \cos^4\theta,$$

which differ for $0 < \theta < \frac{\pi}{2}$. The first measurement creates the coherence (relative to the stored basis) that the second one feels. This is why the paper assigns order effects to the *commuting* condition of the classical sector (Definition 2), and interference and Bell violations to the state's C and non-separability respectively — one signature per sector condition. □

1.2 Proposition 2 (Data-model embedding)

Build $\mathcal{H}$ by recursion: an atomic field over domain D is a single mode $\mathbb{C}^{D\cup\{\perp\}}$; an ordered array of element type τ is the sequence space $\bigoplus_k \mathcal{H}_\tau^{\otimes k}$, and a set/multiset is its symmetric subspace; a nested document is the tensor product of its sub-fields.

Proof. Each record vector $|t\rangle = \bigotimes_f |t_f\rangle$ is by construction a *product vector* across the field factors ($\perp$-padding for heterogeneous collections keeps this true for document data). A relation or collection r is the state $\rho_r = \frac{1}{|r|}\sum_{t\in r}|t\rangle\langle t|$, a convex combination of product states — **separable by the definition of separability**; no positive-partial-transpose or negativity criterion is invoked, so the bound-entanglement counterexamples to "PPT $\Rightarrow$ separable" (which exist from dimension 2×4 and 3×3 up) are irrelevant to the argument. ρ_r is also diagonal in the record basis, so $C = 0$, and selection/membership projectors $\{|t\rangle\langle t|\}$ commute. By Proposition 1 the state lies in the classical sector. Nothing in the argument depends on whether a field factor is a single mode (Codd's 1NF) or an enriched sequence/nesting space, so the relational and document models occupy one and the same diagonal, separable sector. *Remark:* in the paper's 2×2 example $\{(a,x),(b,y)\}$, quoting "negativity 0" *is* conclusive, because PPT $\Leftrightarrow$ separable holds in dimensions 2×2 and 2×3 (Horodecki); the caveat above applies only to higher dimensions. □

1.3 Proposition 3 (Temporal order-invariance)

Proposition 1 (restated)**.** A set of temporal queries over an entity's history is order-invariant — its outcomes admit a single classical joint distribution, every marginal independent of query order — iff the histories' decoherence functional obeys $\operatorname{Re} D(\alpha,\beta) = 0$ for $\alpha \neq \beta$. Logging every intermediate stage the queries distinguish is sufficient; it is necessary exactly when the unlogged alternatives carry coherence.

Proof. The equivalence between the existence of a single classical joint distribution (hence order-independent marginals) and the weak consistency condition $\operatorname{Re} D(\alpha,\beta) = 0$ is the standard consistency theorem of the consistent-histories formalism (Griffiths). For the second clause: a logged intermediate stage inserts a projective measurement at that stage, which dephases the history alternatives and forces $\operatorname{Re} D(\alpha, \beta) = 0$ — sufficiency. Necessity fails in general: a state already diagonal on the alternatives (a classical-sector state, Definition 2 of the paper) satisfies the condition with no logging at all — which is precisely why an ordinary unlogged classical database is order-safe. For states carrying coherence between the alternatives, a generic relative phase gives $\operatorname{Re} D(\alpha, \beta) \neq 0$, and logging is then exactly what restores order-safety. Hence: *once histories carry coherence*, logging granularity is the order-safety boundary. □

1.4 Characterization of the sector: one-way guarantees

The paper deliberately states the characterization as one-way guarantees. Two remarks make precise why the naive "iff" would be false. (i) $|S| \leq 2$ does not imply separability: Werner states in 2×2 are entangled yet satisfy the CHSH bound for all measurement settings (and in a visibility range even admit a full local-hidden-variable model). $S > 2$ *certifies* non-separability; $S \leq 2$ certifies nothing. (ii) For a fixed state, order-invariance of the realised queries does not imply the query algebra commutes. The correct statement quantifies over states (Lemma 1): a commuting family is order-invariant on *every* state, and any observed signature beyond its classical bound certifies that the (state, queries, histories) triple has left the sector.

1.5 CHSH: the two-line classical bound and both ends

If each item carries predetermined ± 1 answers a, a', b, b' under both settings of each side, then $a(b + b') + a'(b - b')$ has $b \pm b' \in \{0, \pm 2\}$, one of the two brackets vanishing, so the expression is ± 2 and $|S| = |\mathbb{E}[\cdot]| \leq 2$. Mixtures (noisy or contextual shared-latent strategies) are convex combinations and inherit the bound. The companion script `qfdb_commoncause_bound.py` enumerates all 16 deterministic common-cause strategies (all lie in $[-2, 2]$), samples 2×10^5 random mixtures (max $|S| = 1.4514$), and evaluates the Bell state with CHSH-optimal settings ($S = 2\sqrt{2} \approx 2.8284$).

2 Worked example: full algebra

Lifecycle: *calf* $\rightarrow$ *steer* $\rightarrow$ *ox* (one male animal; the paper's running example).

Face A (order effect on a stored record). State $|\neg\text{vacc}\rangle$ stored at plane t; mutant-susceptibility query = projector at angle θ. Asked alone: $\Pr(\neg\text{vacc}) = 1.000$. Asked after the mutant query: $\Pr = \cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2}$ in Bloch convention, giving 1.000, 0.875, 0.750, 0.625, 0.500 at $\theta = 0°, 30°, 45°, 60°, 90°$ — an order effect up to 0.500, while the Wang–Busemeyer QQ invariant is 0.000 at every θ (`qfdb_order_effect_mutant.py`).

A classical disturbance model tuned to the same order effect — B-first fully redraws A (50/50), B $\sim$ Bernoulli(r) — gives

$$\text{QQ} = \big[p(A_y B_y) + p(A_n B_n)\big] - \big[p(B_y A_y) + p(B_n A_n)\big] = r - \tfrac{1}{2},$$

i.e. $\text{QQ} = 0.10, 0.20, 0.30$ at prevalence $r = 0.6, 0.7, 0.8$: zero only at one fine-tuned base rate, whereas the quantum value is parameter-free. Pinning that residual is what S1 targets.

Face B (unlogged history). Histories {through-steer, not-through-steer} with amplitudes a, b and relative phase φ. Unlogged: Pr(ox now) $= |a + be^{i\varphi}|^2$ ranges over $[(|a| - |b|)^2, (|a| + |b|)^2]$; the observed swing $[0.000, 0.444]$ forces $|a| = |b|$ with $4|a|^2 = 0.444$, i.e. $|a|^2 = 1/9$. Logged: the decoherence functional is dephased, $\Pr = |a|^2 + |b|^2 = 2/9 = 0.222$, phase-blind. The decoherence functional's off-diagonal magnitude is 0.111 unlogged and 0.000 logged (`qfdb_consistent_histories.py`).

Complementarity (supporting check). With visibility V and which-structure distinguishability D, $V^2 + D^2 \leq 1$ (Englert). Reading records as a which-structure detector: pinning records ($D \to 1$) sharpens the schema; leaving them indefinite keeps it field-like — a quantum form of schema-on-write vs. schema-on-read. The inequality holds for any two-outcome system, classical bits included, so it is a coherence check of the framework, not a non-classicality test.

3 S1 elicitation design (full)

Parties = two category systems describing the same products. Settings: $A = \{$Electronics, Furniture$\}$, $A' = \{$Gift, Tool$\}$; $B = \{$Gadget, Decor$\}$, $B' = \{$Toy, Hardware$\}$. One respondent sees *one* item and *one* setting pair (between-subjects, randomised — the free-choice condition), makes a forced-choice category judgment on each side, coded ± 1. Acceptance criterion: violation iff the 95% bootstrap CI on S lies above 2 *and* the no-signaling (marginal-law) check passes; marginal shifts with the distant setting read as contextuality-with-signaling, not a Bell violation. A violation certifies non-classical structure in correspondence *judgments* (not in the stored schema-field — the Level-2 distinction). Power: for an effect near the Aerts–Sozzo magnitude ($S = 2.42$, $N = 81$), on the order of 10^2 respondents per setting pair; a pre-registered interim look allows early stopping. The Aerts–Sozzo line has been criticised for marginal-selectivity violations (Dzhafarov–Kujala); the no-signaling gate above is the design answer. Instrument, tally and no-signaling scripts: `s1_elicitation/`.

4 Pre-registered stored-data negative control

Pre-registration frozen before evaluation (`s1_stored/PREREGISTRATION.md`, sha256 prefix `72cf065266be0124`). Prediction: stored two-store correlations are common-cause, hence $S \leq 2$ everywhere, adversarial settings included; for per-side-deterministic responses this is a theorem (the shared item is the latent variable of Section 1.5), so the run validates the instrument end-to-end and measures the margin.

Data. Magellan/ditto entity-matching benchmarks; items = ground-truth matched pairs; rule library (numeric-median, length, top-8 token rules) built on TRAIN per side; evaluation on held-out VALID$\cup$TEST. Locality by construction: each side's ± 1 response is a function of that side's record and setting only. **Analyses.** (A) natural settings, same items; (B) adversarial: exhaustive search over setting pairs maximising $|S|$ on TRAIN, evaluated once held-out; (C) between-subjects dry run (four disjoint item groups) with no-signaling deltas. **Self-tests** (pass before data): Bell 2.8284; engineered classical correlations attain $S = 2.000$ exactly (the instrument is sharp at the bound); product state ≈ 0.

Beer (≈ 20 held-out matches) is excluded by a ≥ 30-item practicality rule adopted at analysis time and disclosed here (it was not part of the frozen pre-registration); all other pairs are reported without selection. The between-subjects no-signaling deltas scale with per-group n —

Table 1: Held-out results (bootstrap 95% CIs, 10^4 resamples). No CI lies above 2: the classical-sector prediction is verified on every analysable pair, while the same pipeline's entangled control reaches 2.8284.

dataset	n	S_{nat}	$S_{\text{adv}}^{\text{train}}$	$S_{\text{adv}}^{\text{held-out}}$	95% CI	NS Δ
Amazon–Google	468	$+1.026$	$+1.966$	$+1.974$	$[+1.940, +2.000]$	0.154
DBLP–ACM	888	-0.189	$+2.000$	$+1.995$	$[+1.986, +2.000]$	0.189
DBLP–GoogleScholar	2140	-0.456	$+1.988$	$+1.983$	$[+1.972, +1.993]$	0.037
Fodors–Zagats	44	-0.364	$+2.000$	$+2.000$	$[+2.000, +2.000]$	0.364
Walmart–Amazon	386	-0.166	$+1.979$	$+1.938$	$[+1.886, +1.979]$	0.103
iTunes–Amazon	54	$+0.815$	$+2.000$	$+2.000$	$[+2.000, +2.000]$	0.319

the direct lesson for the human study's sample size. Adversarial search saturates the deterministic bound on strongly-correlated pairs (exactly 2.000) and crowds it elsewhere (1.94–1.99): real integration data pushes against the classical ceiling but cannot pass it, exactly as the embedding requires. Reproduce with `s1_stored/s1_stored_negative_control.py` (fetch commands in the pre-registration).

5 LLM-respondent pilot harness

`s1_elicitation/llm_harness/` implements the S1 protocol with a language model as synthetic respondent: one API call = one respondent (fresh context; between-subjects), one setting pair per call, neutral catalogue-filing wording (no physics vocabulary), option order counterbalanced, temperature > 0 with a reported grid, open-weights model preferred and version pinned, full JSONL logs, and the same acceptance criterion (CI > 2 *and* no-signaling). A violation would certify non-classical structure in the *model's* judgment statistics — not in human cognition and not in the schema-field; the pilot's roles are instrument validation and an effect-size prior for the human study. A `--dry-run` classical simulated judge exercises the analysis path without any API.

6 Reproducibility

Reference environments. All results in this report were reproduced digit-for-digit (including bootstrap CIs, via a fixed PCG64 seed) on two independent machines: (a) a MacBook Pro (Apple M4, 24 GB RAM, 512 GB internal SSD + 1 TB Thunderbolt external SSD) running macOS Tahoe 26.x with NumPy 2.2.6, and (b) a Debian Linux aarch64 container with Python 3.10 and NumPy 2.2.6. No GPU or quantum hardware is used; the full negative-control run completes in under a minute on either machine.

All scripts are NumPy-only (the LLM harness is stdlib-only). `qfdb_signatures_verify.py`, `qfdb_order_effect_mutant.py`, `qfdb_consistent_histories.py`, `qfdb_defs_verify.py`, `qfdb_commoncause_bo` `qfdb_H1_field_modes.py` reproduce every computed number in the main paper; `s2_calibration/` reproduces the 72-survey calibration (order-effect mean 0.087, range 0.03–0.58; QQ mean $+0.001 \pm 0.036$, $t(71) = 0.25$, $p = 0.80$); `s1_stored/` reproduces the table above.